CLOUD COMPUTING LAB MANUAL

Subject: Cloud Computing (BCS601)

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Experiment 1: VM Creation and AWS CloudShell

Aim

Create and manage EC2 virtual machine using AWS CloudShell.

Procedure

1. Login to AWS Console.
2. Use AWS Search Bar to open the required service.
3. Click Create or Launch option.
4. Enter required configuration details.
5. Select default settings wherever applicable.
6. Configure security and permissions.
7. Create required resources.
8. Run commands or upload files if needed.
9. Verify output and execution status.
10. Delete resources after completion.
11. Search EC2 service.
12. Click Launch Instance.
13. Select Amazon Linux AMI.
14. Choose t2.micro instance type.
15. Create Key Pair and download PEM file.
16. Allow SSH traffic on Port 22.
17. Launch EC2 instance.
18. Connect using EC2 Instance Connect.

Result

Successfully executed experiment 1: vm creation and aws cloudshell.

Viva Questions

1. **What is the main purpose of this experiment?**
   - It helps understand AWS cloud services practically.
2. **What AWS service was used?**
   - AWS managed cloud services were used for execution.
3. **What is scalability?**
   - Scalability increases resources based on workload.
4. **What is cloud security?**
   - Cloud security protects resources and data.

Experiment 2: Amazon S3 and CloudShell

Aim

Create and manage Amazon S3 bucket and objects.

Procedure

1. Login to AWS Console.
2. Use AWS Search Bar to open the required service.
3. Click Create or Launch option.
4. Enter required configuration details.
5. Select default settings wherever applicable.
6. Configure security and permissions.
7. Create required resources.
8. Run commands or upload files if needed.
9. Verify output and execution status.
10. Delete resources after completion.
11. Search Amazon S3.
12. Click Create Bucket.
13. Upload files into bucket.
14. Enable static website hosting.
15. Copy object URL and test output.

Result

Successfully executed experiment 2: amazon s3 and cloudshell.

Viva Questions

1. **What is the main purpose of this experiment?**
   - It helps understand AWS cloud services practically.
2. **What AWS service was used?**
   - AWS managed cloud services were used for execution.
3. **What is scalability?**
   - Scalability increases resources based on workload.
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   - Cloud security protects resources and data.

Experiment 3: DynamoDB and Cloud Functions

Aim

Create DynamoDB table and execute cloud functions.

Procedure

1. Login to AWS Console.
2. Use AWS Search Bar to open the required service.
3. Click Create or Launch option.
4. Enter required configuration details.
5. Select default settings wherever applicable.
6. Configure security and permissions.
7. Create required resources.
8. Run commands or upload files if needed.
9. Verify output and execution status.
10. Delete resources after completion.
11. Search DynamoDB.
12. Create new table with partition key.
13. Insert sample items into table.
14. Create cloud function and test execution.

Result

Successfully executed experiment 3: dynamodb and cloud functions.

Viva Questions

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   - It helps understand AWS cloud services practically.
2. **What AWS service was used?**
   - AWS managed cloud services were used for execution.

3. **What is scalability?**
   - Scalability increases resources based on workload.
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   - Cloud security protects resources and data.

Experiment 4: API Gateway and App Engine

Aim

Create REST API using API Gateway.

Procedure

1. Login to AWS Console.
2. Use AWS Search Bar to open the required service.
3. Click Create or Launch option.
4. Enter required configuration details.
5. Select default settings wherever applicable.
6. Configure security and permissions.
7. Create required resources.
8. Run commands or upload files if needed.
9. Verify output and execution status.
10. Delete resources after completion.
11. Search API Gateway.
12. Create REST API.
13. Create GET and POST methods.
14. Deploy API and test endpoint.

Result

Successfully executed experiment 4: api gateway and app engine.

Viva Questions

1. **What is the main purpose of this experiment?**
   - It helps understand AWS cloud services practically.
2. **What AWS service was used?**
   - AWS managed cloud services were used for execution.

3. **What is scalability?**

   - Scalability increases resources based on workload.

4. **What is cloud security?**

   - Cloud security protects resources and data.

Experiment 5: AWS Lambda and Cloud Storage

Aim

Create Lambda function integrated with cloud storage.

Procedure

1. Login to AWS Console.
2. Use AWS Search Bar to open the required service.
3. Click Create or Launch option.
4. Enter required configuration details.
5. Select default settings wherever applicable.
6. Configure security and permissions.
7. Create required resources.
8. Run commands or upload files if needed.
9. Verify output and execution status.
10. Delete resources after completion.
11. Search Lambda service.
12. Create Lambda function.
13. Write sample Python code.
14. Create trigger using S3 event.

Result

Successfully executed experiment 5: aws lambda and cloud storage.

Viva Questions

1. **What is the main purpose of this experiment?**
   - It helps understand AWS cloud services practically.
2. **What AWS service was used?**
   - AWS managed cloud services were used for execution.

3. **What is scalability?**
   - Scalability increases resources based on workload.
4. **What is cloud security?**
   - Cloud security protects resources and data.

Experiment 6: Docker and Cloud SQL

Aim

Deploy Docker container and connect Cloud SQL.

Procedure

1. Login to AWS Console.
2. Use AWS Search Bar to open the required service.
3. Click Create or Launch option.
4. Enter required configuration details.
5. Select default settings wherever applicable.
6. Configure security and permissions.
7. Create required resources.
8. Run commands or upload files if needed.
9. Verify output and execution status.
10. Delete resources after completion.
11. Launch EC2 instance.
12. Install Docker using yum command.
13. Pull Docker image from Docker Hub.
14. Run container and verify output.

Result

Successfully executed experiment 6: docker and cloud sql.

Viva Questions

1. **What is the main purpose of this experiment?**
    - It helps understand AWS cloud services practically.
2. **What AWS service was used?**
    - AWS managed cloud services were used for execution.

3. **What is scalability?**
    - Scalability increases resources based on workload.
4. **What is cloud security?**
    - Cloud security protects resources and data.

Experiment 7: Redis and ElastiCache

Aim

Configure Redis caching using ElastiCache.

Procedure

1. Login to AWS Console.
2. Use AWS Search Bar to open the required service.
3. Click Create or Launch option.
4. Enter required configuration details.
5. Select default settings wherever applicable.
6. Configure security and permissions.
7. Create required resources.
8. Run commands or upload files if needed.
9. Verify output and execution status.
10. Delete resources after completion.
11. Search ElastiCache.
12. Create Redis cluster.
13. Configure cache node settings.
14. Test caching operations.

Result

Successfully executed experiment 7: redis and elasticache.

Viva Questions

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   - It helps understand AWS cloud services practically.
2. **What AWS service was used?**
   - AWS managed cloud services were used for execution.

3. **What is scalability?**
   - Scalability increases resources based on workload.
4. **What is cloud security?**
   - Cloud security protects resources and data.

Experiment 8: VPC and CloudFront

Aim

Create VPC network and configure CloudFront.

Procedure

1. Login to AWS Console.
2. Use AWS Search Bar to open the required service.
3. Click Create or Launch option.
4. Enter required configuration details.
5. Select default settings wherever applicable.
6. Configure security and permissions.
7. Create required resources.
8. Run commands or upload files if needed.
9. Verify output and execution status.
10. Delete resources after completion.
11. Search VPC service.
12. Create VPC and subnet.
13. Attach Internet Gateway.
14. Create CloudFront distribution.

Result

Successfully executed experiment 8: vpc and cloudfront.

Viva Questions

1. **What is the main purpose of this experiment?**
   - It helps understand AWS cloud services practically.
2. **What AWS service was used?**
   - AWS managed cloud services were used for execution.

3. **What is scalability?**
    - Scalability increases resources based on workload.
4. **What is cloud security?**
    - Cloud security protects resources and data.

Experiment 9: AWS Step Functions and Monitoring

Aim

Create workflow using Step Functions.

Procedure

1. Login to AWS Console.
2. Use AWS Search Bar to open the required service.
3. Click Create or Launch option.
4. Enter required configuration details.
5. Select default settings wherever applicable.
6. Configure security and permissions.
7. Create required resources.
8. Run commands or upload files if needed.
9. Verify output and execution status.
10. Delete resources after completion.
11. Search Step Functions.
12. Create State Machine.
13. Paste workflow JSON code.
14. Start workflow execution.

Result

Successfully executed experiment 9: aws step functions and monitoring.

Viva Questions

1. **What is the main purpose of this experiment?**
   - It helps understand AWS cloud services practically.
2. **What AWS service was used?**
   - AWS managed cloud services were used for execution.

3. **What is scalability?**
   - Scalability increases resources based on workload.
4. **What is cloud security?**
   - Cloud security protects resources and data.

Experiment 10: Kubernetes and CI/CD

Aim

Create CI/CD pipeline and Kubernetes deployment.

Procedure

1. Login to AWS Console.
2. Use AWS Search Bar to open the required service.
3. Click Create or Launch option.
4. Enter required configuration details.
5. Select default settings wherever applicable.
6. Configure security and permissions.
7. Create required resources.
8. Run commands or upload files if needed.
9. Verify output and execution status.
10. Delete resources after completion.
11. Search CodePipeline.
12. Create CI/CD pipeline.
13. Connect GitHub repository.
14. Deploy application using CodeDeploy.

Result

Successfully executed experiment 10: kubernetes and ci/cd.

Viva Questions

1. **What is the main purpose of this experiment?**
   - It helps understand AWS cloud services practically.
2. **What AWS service was used?**
   - AWS managed cloud services were used for execution.

3. **What is scalability?**
    - Scalability increases resources based on workload.
4. **What is cloud security?**
    - Cloud security protects resources and data.